

Reese Ludwick

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EDUCATION

Bachelor of Science, Computer Science — Shepherd University (Aug 2021 – May 2025), Major GPA 3.43

EXPERIENCE

Software Associate

Remote

Cognizant Technology Solutions

Dec 2025 – Present

- Develop and maintain GenAI and agentic AI systems using LangChain, LangGraph, and Python across R&D initiatives and client-facing deliverables.
- Design and implement RAG pipelines and MCP server integrations connecting enterprise tools and language models for production-ready workflows.
- Collaborate cross-functionally to scope, prototype, and deliver AI-powered data analysis tools that drive measurable business insights.

Freelance AI Engineer & Full-Stack Developer

Remote

Independent Contract, WorldCovers / APMC

May 2026 – Present

- Paid contract engineer maintaining production web properties (chinaoverprints.com, ianthom.org) and building WorldCovers, a philatelic cataloging platform sponsored by the U.S. Philatelic Classics Society.
- Deliver features and automation against a 14-table PostgreSQL schema powering a Django REST API backend with a React / TypeScript frontend.

GenAI Developer Contractor

Remote

Cognizant Technology Solutions

Jun 2025 – Dec 2025

- Built custom FastMCP servers and deployed external MCP integrations alongside AWS infrastructure to give agents dynamic tool access in production-style agentic workflows.
- Built and delivered GenAI proof-of-concept solutions with LangChain, LangGraph, and Python; applied prompt engineering and model evaluation to optimize LLM outputs.

IT Student Helper

Hedgesville, WV

James Rumsey Technical Institute

Aug 2019 – May 2021

- Resolved 300+ support tickets; contributed to a campus Wi-Fi overhaul that cut latency 25%; configured and tested 100+ AV and networking systems.

PROJECTS

CreatorClip / autoclip.studio — *Python, FastAPI, Celery, Deepgram, pgvector, Voyage AI, Anthropic API, ffmpeg, React/TypeScript, YouTube OAuth*

- Live, multi-tenant production SaaS (autoclip.studio) with Stripe billing and a React/TypeScript frontend — turns long-form YouTube content into short-form clips, learning a creator's channel style via a custom "DNA" module.
- Async Celery pipeline: Deepgram transcription (WhisperX/AssemblyAI selectable) → Voyage AI + pgvector embeddings → ffmpeg rendering, with full YouTube OAuth publishing.

CFO Agent — *Python, FastAPI, LangGraph, PostgreSQL, Redis, Anthropic API*

- Personal-finance AI agent (in active development) on a FastAPI + LangGraph backend with PostgreSQL persistence and Redis caching.
- Hybrid context architecture over a Fernet-encrypted data vault with CSV/OFX import; architected for multi-tenant productization.

Agentic Software Development Team — *Python, LangChain, LangGraph, Amazon Bedrock (Cognizant R&D)*

- Multi-agent LangGraph system with specialized agents for code generation, testing, security review, and supervision — automating the full software build lifecycle end to end.
- Integrated Amazon Bedrock and open-source ML models as swappable inference backends.

Customer Call Data Analysis & Insights Engine — *Python, NLP, Sentence Transformers (Cognizant R&D)*

- NLP clustering with sentence-transformer embeddings grouping large volumes of customer call data into semantic issue segments; agentic analysis feeding executive reporting.

TECHNICAL SKILLS

Languages: Python, SQL

AI / Agentic: LangChain, LangGraph, RAG pipelines, MCP servers (FastMCP), Amazon Bedrock, Anthropic API, Sentence Transformers, TensorFlow, pgvector, prompt engineering, model evaluation

Backend / Infra: FastAPI, Celery, Django REST, PostgreSQL, Redis, Docker

Frontend / Tooling / Cloud: React / TypeScript, AWS, Azure Fundamentals, Linux/Ubuntu, Git/GitHub, VS Code, Claude Code, Jupyter

CERTIFICATIONS

TensorFlow Developer Specialization (DeepLearning.AI) · Microsoft Certified: Azure Fundamentals (AZ-900) · Linux + Git (Boot.dev) · GitHub Copilot, Telecom & Generative AI Context Engineering (Cognizant)